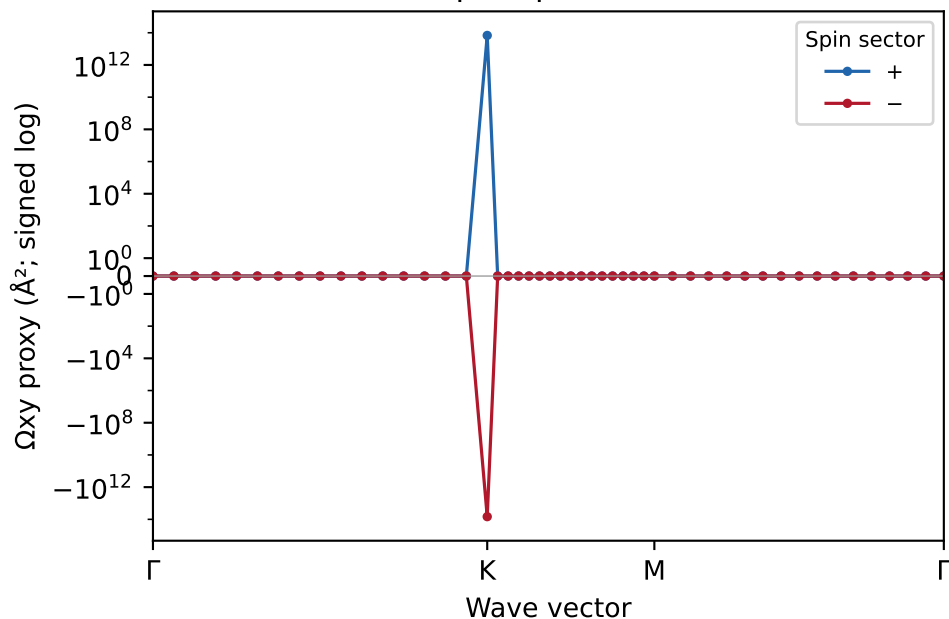


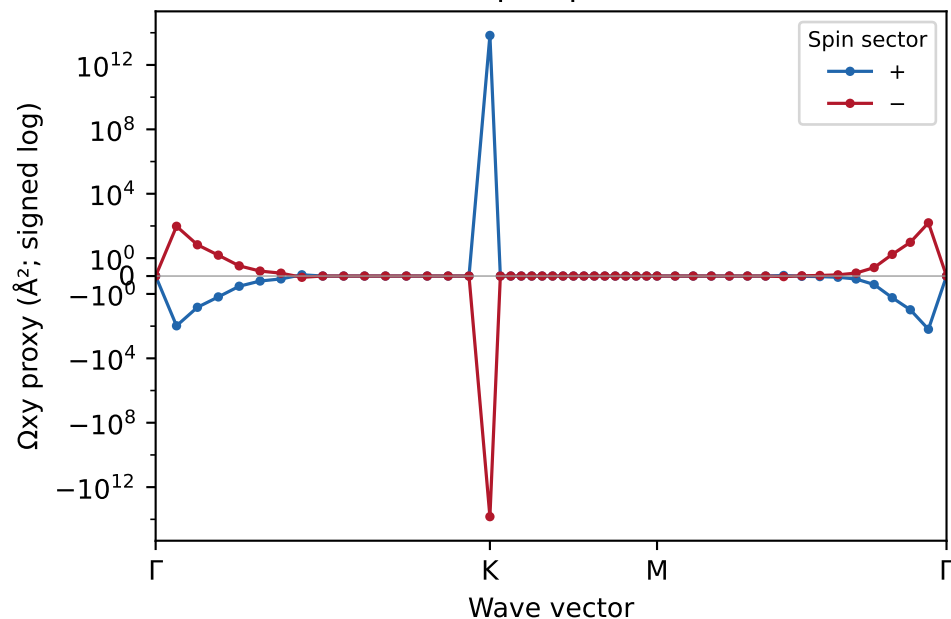
Graphene: occupied and selected-pair spin sectors

Canonical-momentum derivative proxy · pseudo Gram metric · no integer rounding

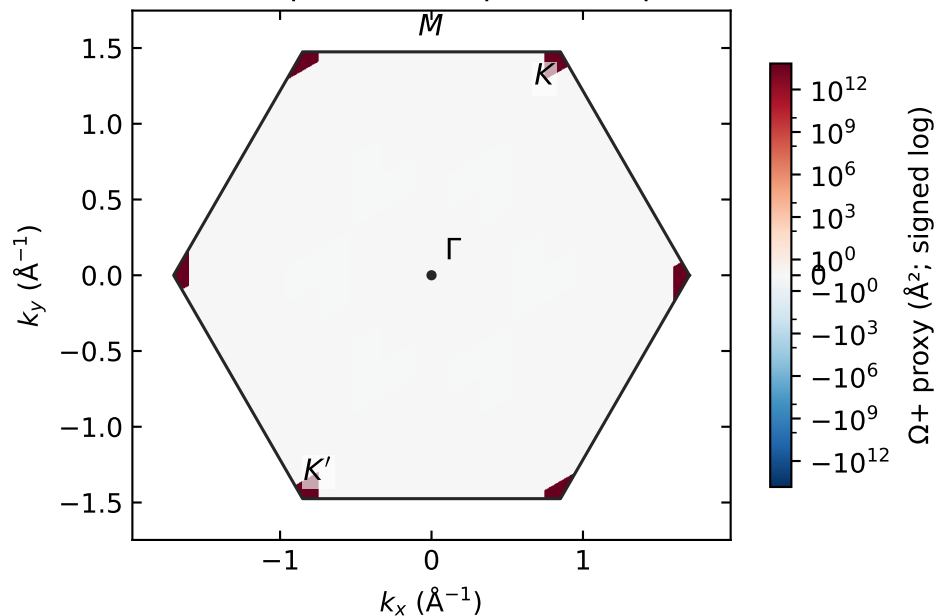
(a) Occupied path bands 1:8



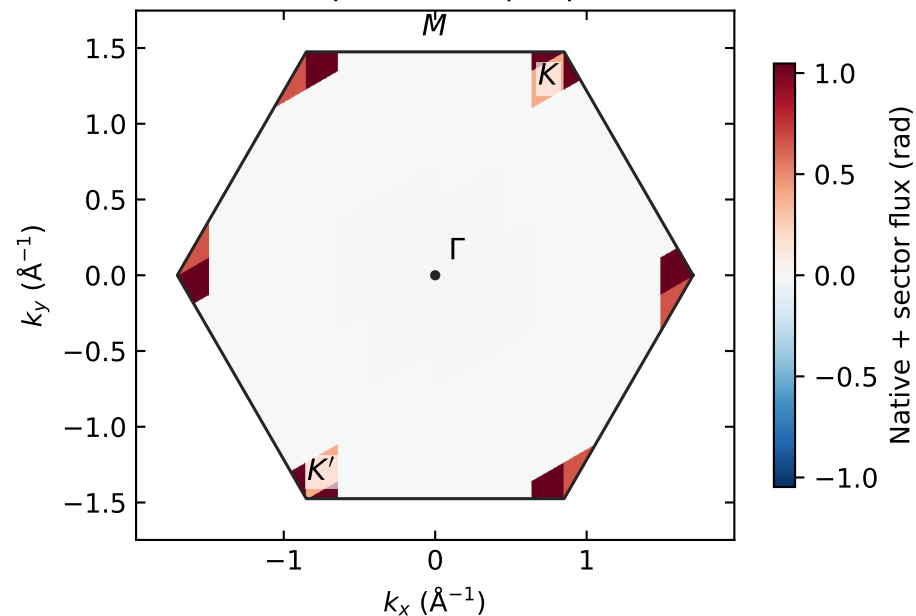
(b) Selected pair path bands 7:8



(c) Occupied full-BZ point samples



(d) Occupied Fukui plaquettes



Subspace	Raw C_est, +	Raw C_est, -	Fukui C+	Fukui C-
occupied 1:8	1.135907e+12	-1.135907e+12	1	-1
pair 7:8	1.135907e+12	-1.135907e+12	unassigned	unassigned

Lines connect stored points; cell colors are constant. C_est is not a certified topological invariant.